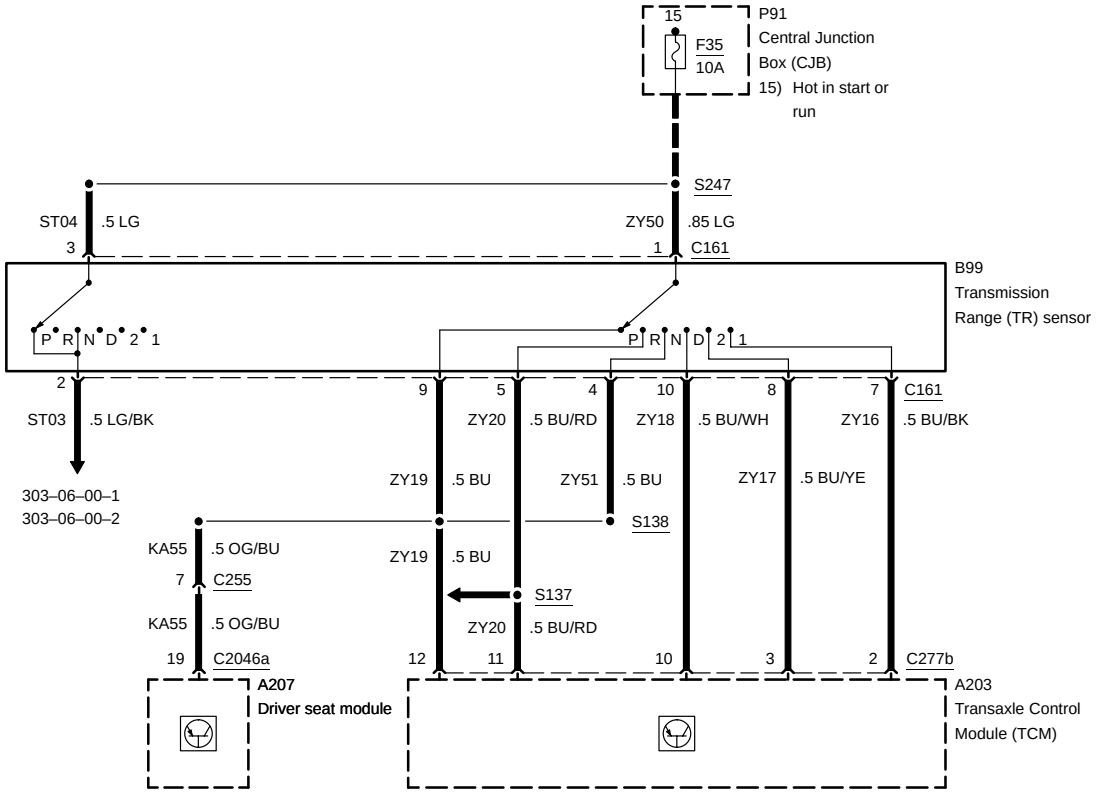with memory -5